Intro to NLP for corpus linguistics

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University of Hawaii at Manoa

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Topics to cover

- Course logistics
- Intro to course
- Intro to NLP
- Intro to corpus linguistics
- Intro to NLP and language learning

Course logistics

Course logistics

- **Instructor**: Hakyung Sung (Prefer to be called Hakyung; she/her)
  - PhD in Linguistics, MS in Computer Science @ University of Oregon
  - Study how second language learners use words and grammatical structures, using natural language processing approaches
- **Time:** W 3:00-5:50 PM
- **Office hours:**
  - M 1:00-2:00 PM (in-person, Moore 554)
  - By appointment (Email me)
- **Course website:** `https://hksung.github.io/Fall26_SLS775/`
  - Link on the Lamakū
- **Email:** `sungh@hawaii.edu`

Grading

- **Syllabus includes all the details/updates (which is on the course website).**

- (1) Presentation 10%

  https://hksung.github.io/Fall26_SLS775/assignments/1_paper%20presentation

Tentative course schedule

Week	Date	Topic
1	8/26	Intro to NLP for corpus linguistics
		Lab: Intro to Python 1
2	9/2	NLP in learner corpus research Presentation 1 (Kyle & Eguchi, 2024)
		Lab: Intro to Python 2
3	9/9	Corpus design and selection Presentation 2 (Hanks et al., 2024)
		Lab: A survey of corpora

Grading

- (2) Assignments 4 x 10%

Week	Date	Topic	Readings	Assignment Due Friday, 11:59 pm
1	8/26	Intro to NLP for corpus linguistics **Lab:** Intro to Python 1	Dunn (2022) Gries (2026) Meurers (2020)	Survey
2	9/2	NLP in learner corpus research Presentation 1 (Kyle & Eguchi, 2024) **Lab:** Intro to Python 2	Granger (2026) Kyle (2021) Kyle & Sung (2026)	
3	9/9	Corpus design and selection Presentation 2 (Hanks et al., 2024) **Lab:** A survey of corpora	Gries (2009) Egbert et al. (2022), Ch. 3	Assignment 1

Grading

- (3) **Project** 40%
  - Proposal 3%
  - Midway presentation 5%
  - Midway report 7%
  - Final presentation 10%
  - Final paper 15%

Grading

- (4) Participation 10%
    - Completing ad-hoc events that the instructor announces in class 2%
    - Active participation in Presentation 4%
    - Active participation in midway/final presentation 4%

Grading

Extra credit 2%

You may participate in the Linguistics Beyond the Classroom (LBC) program for an extra 2% credit towards your final course grade. The purpose of this program is for students to get first-hand experience of language research by participating in an on-going research project conducted here at UH. For more information, see http://ling.lll.hawaii.edu/sites/lbc/. You may complete either Option 1 (research participation) or Option 2 (500-word event reflection) as outlined on the LBC website. I must receive your completed questionnaire (option 1) or event response (option 2) by December 4th, 2026 for you to receive this credit.

Other policies

All the information is available in the syllabus:

- Grading policy (including CR/NC; *Audit)
- Attendance policy
- Late policy
- Use of AIs
- UH policies and resources

Expectation

- Ideally, read the assigned readings before/after class.
  - My summary is in the lecture notes.
  - My lecture is built on the lecture notes.
- Be present.
- Bring your laptop.
  - Lecture $\rightarrow$
  - Presentation $\rightarrow$
  - Lab

Expectation

Generated by ChatGPT

Intro to course

Course title

Natural Language Processing in Corpus Research

Course title

Natural Language Processing in **Corpus Research**

What counts as corpus research?

Givón (2009)

Task: Analyzing their discourse contexts

> *"What I have done with the [...] CHILDES transcripts [...] is isolate and extract all the MIUs [...] that surround each complex VP construction."*

(3) **Simple modal interactions**

a. **Deontic**: (Eve-I, p. 2)

EVE: Napkin.	(request)
MOT: Oh, do you **want** a napkin too?	(offer)

b. **Deontic**: (Eve-I, p. p. 3)

EVE: Fraser blow nose, blow nose.	(request)
MOT: Wipe your nose? **Can** you blow?	(offer)

c. **Deontic**: (Eve-I, p. 15-16)

EVE: Bottle.	(request)
MOT: What?	(request for interpretation)
EVE: Eve...	(request)
MOT: Do you **want** to taste it?	(offer)
Let's see if Sarah **would like** to **have** a drink	(manipulation)
EVE: Eve **want** some too.	(request)

Is this corpus research?

- Why or why not?
- What makes something a *corpus*?
- What makes a study *corpus research*?

Givón, T. (2009). *The genesis of syntactic complexity: Diachrony, ontogeny, neuro-cognition, evolution*. John Benjamins.

What counts as corpus research?

Corpus

1. ~~the body of a person or animal, especially when dead~~
2. ~~*Anatomy*. a body, mass, or part having a special character or function~~
3. a large or complete collection of writings
4. *Linguistics*. a body of utterances, as words or sentences, assumed to be representative of and used for lexical, grammatical, or other linguistic analysis

`https://www.dictionary.com/browse/corpus`

Is this corpus research?

- Why or why not?
- What makes something a *corpus*?
- What makes a study *corpus research*?

So, what is corpus research?

In a broad sense, if you analyze a **body of utterances** for linguistic purposes, you are doing **corpus research**.

- Large collection? How large it should be?
- Researchers assume that the collection is meaningful and appropriate ($\approx$ *representative*; will cover this in Week 3) for the linguistic question being investigated.
- NLP is therefore NOT what makes research corpus research.

Course title

Natural Language Processing (NLP) in Corpus Research

Example

Givón (2009) Task: Analyzing their discourse contexts

(3) **Simple modal interactions**

a. **Deontic**: (Eve-I, p. 2)

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Is this an NLP approach?

Givón, T. (2009). *The genesis of syntactic complexity: Diachrony, ontogeny, neuro-cognition, evolution*. John Benjamins.

Givón (2009) Task: Analyzing their discourse contexts

(3) **Simple modal interactions**

a. **Deontic**: (Eve-I, p. 2)

EVE: Napkin.	(request)
MOT: Oh, do you **want** a napkin too?	(offer)

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EVE: Eve **want** some too.	(request)

- Can we replicate the analysis?
- Can we apply the analysis consistently across datasets?
- Can we scale it up?

- Can NLP make these analyses more systematic and reproducible?

Givón, T. (2009). *The genesis of syntactic complexity: Diachrony, ontogeny, neuro-cognition, evolution*. John Benjamins.

Corpus linguistics and NLP

Corpus Linguistics

- Begins with a linguistic question
- Examines naturally occurring language
- Describes patterns of language use (e.g., frequency, keyness, association strength)
- Interprets linguistic distributions

+NLP

- Develops computational methods
- Automatically processes language
- Identifies and predicts linguistic properties
- Replicates findings both within/outside the research group (to argue that your findings were not serendipity but part of science)

NLP for corpus linguistics

Linguistic question

↓

Corpus data

↓

Computational analysis

↓

Evidence about language use

↓

Answer question

Two broad computational problems (Dunn, 2022)

Categorization

What **kind** of thing is this?

- Part-of-speech category
- Grammatical construction
- Register / genre
- Error type
- Proficiency level

Comparison

How **similar or related** are these?

- Words
- Sentences
- Texts
- Speakers
- Registers

But the distinction is not about algorithms

$$\text{Categorization} \neq \text{supervised learning}$$

$$\text{Comparison} \neq \text{unsupervised learning}$$

- **Supervised learning**: learns from examples with predefined target labels.
- **Unsupervised learning**: identifies patterns or groupings without predefined target labels.
- will cover in Week 9 & 11

The same problem may be approached in multiple computational ways.

▸ Example 1 ▸ Example 2

Intro to NLP

What is NLP?

- NLP
- machine learning
- deep learning
- LLM
- AI
- ...

What is NLP?

Manning, 2022, pp. 127-128

> *"When scientists consider* ***artificial intelligence****, they mostly think of modeling or recreating the capabilities of an* ***individual human brain****... The power of* ***language*** *is fundamental to human societal intelligence, and language will retain an important role in a future world in which human abilities are augmented by artificial intelligence (AI) tools ...* ***For these reasons, the field of natural language processing (NLP) emerged in tandem with the earliest development in AI****."*

Manning, C. D. (2022). Human language understanding & reasoning. Daedalus, 151(2), 127-138.

What is NLP?

“NLP is a subfield of ***computer science*** *and artificial intelligence (AI) that uses* ***machine learning*** *to enable computers to understand and communicate with human language.”*

https://www.ibm.com/think/topics/natural-language-processing

What is NLP?

- Now, people in many fields are so excited about NLP because they:
  - can extract patterns, meanings, and structures that are difficult to capture manually $\rightarrow$ **Major focus**
  - leverage nlp-techniques to generate and explore linguistic hypotheses $\rightarrow$ **Minor focus**
  - enables real-world applications (e.g., search, translation, chatbots, assessment) $\rightarrow$ NOT our focus

What can NLP do? (Generally)

Word level

- Tokenization
- Lemmatization

Sentence level

- Parsing
- Error detection
- Semantic representation
- Translation

Text level

- Classification
- Summarization

What can NLP do? (This class)

Word level

- Tokenization
  Week 4
- Lemmatization
  Week 4
- POS tagging
  Week 6
- Word embeddings
  Week 5

Sentence level

- Dependency parsing
  Week 7
- Linguistic annotation
  Week 9
- Semantic role labeling
  Week 11

Text / discourse level

- Linguistic complexity
  Week 8
- Sentiment / discourse analysis
  Week 13
- Multilingual / multimodal corpora
  Week 14

Why NLP matters for corpus linguistics

NLP transforms unstructured language into representations that can be systematically searched, counted, compared, and modeled.

Why NLP **may NOT matter** for corpus linguistics

NLP does not guarantee validity.

The validity of an NLP-based corpus analysis depends on:

- the accuracy of the computational method;
- the match between the computational output and the linguistic construct of interest.
    - We want to measure *writing ability*.
    - We might use *typing speed* as a measure.
    - Typing speed can be measured accurately.
    - Does it represent writing ability?

Computational output is not automatically linguistically valid.

NLP has become **SO** widespread

- More accessible NLP tools and APIs
- Open-source datasets and standardized benchmarks
- More datasets and models for languages beyond English

It is easy to build an NLP pipeline and generate results from ~~corpus~~ data. data vs. corpus

Again, computational output is not equal to linguistically valid output!

Intro to corpus linguistics

What is corpus linguistics? (Gries, 2026)

Corpus distributions can provide quantitative measures of linguistic experience.

- **Frequency**: how often a form or pattern occurs
- **Co-occurrence**: how often linguistic elements occur together
- **Contingency**: how strongly one element predicts another
- These measures can operationalize constructs such as entrenchment and form–meaning associations.

What is corpus linguistics? (Gries, 2026)

Advantages

- Captures distributions in naturally occurring language
- Can complement controlled experimental evidence (+psycholinguistics)

What is corpus linguistics? (Gries, 2026)

Challenges

- Noisy, heterogeneous, and highly intercorrelated data
- Strongly imbalanced/Zipfian distributions
- Results depend on corpus composition, sampling, and dispersion

Intro to NLP and language learning

NLP and language learning (Meurers, 2020)

NLP can be used to analyze language from two different perspectives:

Language by learners

- What can NLP tell us about **learner language**?
- e.g., proficiency, errors, development, complexity, cross-linguistic influence

Language for learners

- How can NLP help us analyze or adapt **language presented to learners**?
- e.g., readability, vocabulary, materials, exercises, assessment items

What NLP task or measure could you use for each question?

NLP and language learning

Language by learners

- What can NLP tell us about learner language?
- e.g., proficiency, errors, development, complexity, cross-linguistic influence

Language for learners

- How can NLP help us analyze or adapt language presented to learners?

We'll explore this in more detail next week (Week 2).

For the course project

Begin with your questions about language.

Linguistic question

↓

Corpus

↓

NLP method

↓

Interpretation

Please complete the **first-class survey**.

Appendix

Example: Categorization

Suppose we want to classify texts by register.

- *Supervised*: Train a model using texts already labeled as academic writing, news, or conversation.
- *Unsupervised*: Cluster texts according to their linguistic similarities and examine whether the clusters correspond to recognizable registers.

Example: Comparison

How similar are the meanings of two sentences?

- Represent sentences as vectors and calculate *cosine similarity*.
- Or train a supervised model using sentence pairs labeled for degrees of semantic similarity.

Research problem and computational method are not the same thing.

Data $\neq$ Corpus (Gries, 2026)

Language data

- Any collection of linguistic observations
- May or may not have been systematically sampled
- May have limited documentation

Corpus

- Deliberately constructed
- Systematically organized
- Documented
- Designed to support analysis of a particular population or domain